

Ensembling machine learning models to identify forest fire-susceptible zones in Northeast India

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ABSTRACT

Forest fires pose significant challenges by disrupting ecological balance, impacting socio-economic harmony, and raising global concerns. North-East India (NEI) experiences high incidences of forest fires, making it crucial to implement suitable management measures considering the driving forces influencing fire likelihood. This study aims to identify forest fire susceptibility zones in NEI by using five machine-learning modeling approaches, Boosted Regression Tree (BRT), Random Forest (RF), Support Vector Machine (SVM), Classification and Regression Tree (CART), and Multivariate Adaptive Regression Splines (MARS), and an ensemble method. Forest fire data from the SNPP – VIIRS sensor (2018–2019) were rectified for spatial autocorrelation. Thirty-two responsive predictor variables related to topographic, climatic, biophysical, and anthropogenic factors were used as model inputs and multicollinearity analysis was performed to eliminate highly correlated predictors. Results indicate that the southern and southeastern regions of NEI, characterized by ample solar radiation, enhanced vegetation index, high human population density, and *jhum* cultivation, contribute significantly to higher susceptibility to forest fires. The Random Forest model performs best among the models employed, achieving an AUC value of 0.87. The ensemble susceptibility map, binarized based on AUC weighting, covers 29.54% of the total geographic area and 44.42% of the forested area of NEI. The vulnerability levels vary among states, with Mizoram showing the highest susceptibility at 89.27% and Sikkim exhibiting the lowest vulnerability at only 0.49% of their respective geographic areas. This map provides valuable insights for implementing effective forest fire management plans in the region. Moreover, the methodology utilized in this study, which incorporates satellite imagery, GIS techniques, and improved modeling techniques, can be replicated in any geographical region worldwide to facilitate effective forest fire management at a regional to large scale.

1. Introduction

Forests, as diverse ecosystems (Keenan et al., 2015; Morales-Hidalgo et al., 2015) have crucial roles in conserving soil and water, regulating climate, facilitating the carbon cycle, and contributing to various other ecological functions (Bergeron et al., 2004). In this context, forest fire is an important natural phenomenon that acts as an environmental mesh, chooses certain species and features, and drives ecological succession to shape the ecosystem (Parashar and Biswas, 2003; Satendra and Kaushik, 2014; Sultan and Pillai, 2023). Nevertheless, excessive and frequent fires can potentially move beyond their historical boundaries, leading to irreversible alterations in the landscape and its ecological processes (Cha et al., 2020). In recent decades, a significant increase in forest fire

incidents has destroyed large forested land yearly (FAO, 2020; Bui et al., 2018). That has significant global ecological and socio-economic impacts, affecting the environment, infrastructure, and human well-being (Zheng et al., 2021). The University of Maryland's data reveal that forest fires ravaged approximately 27.9 million ha of forests in 2016, and this destructive trend continues to escalate annually (Weisse and Goldman, 2017). Global fires significantly influence the global carbon cycle, contributing to approximately two gigatons (Gt) C/ year of immediate direct carbon emissions to the atmosphere while also influencing the legacy carbon sinks (Van Der Werf et al., 2017).

India is one of the top ten countries worldwide with a significant forest cover, encompassing >72.16 million hectares. This accounts for approximately 2% of the global forest area (Mohanty and Mithal, 2022).

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Around 36% of India's forested regions are susceptible to wildfires, as stated by the Indian State of Forest Report (ISFR, 2019). Compared to other South Asian countries, India ranks as the second most vulnerable nation to forest fires, with a susceptibility rate of 32% (Reddy et al., 2019). Over the past two decades, India has witnessed a drastic increase in intense forest fire incidents, with a staggering rise of 52% (Mohanty and Mithal, 2022) that directly contribute to black carbon emissions and alter the surface albedo, increasing snow melt from the higher Himalayan ranges (Murthy et al., 2019).

Growing concerns about forest fires have prompted increased interest in early warning systems, incorporating advanced technologies like infrared and thermal cameras, video analysis, artificial intelligence, and sensors. This surge in research, particularly within forest fire management organizations, emphasizes the importance of developing effective strategies, including the creation of fire risk maps. These maps help identify vulnerable areas and guide proactive measures to mitigate fire risk and ensure forest sustainability (Sivrikaya and Küçük, 2022). The scientific community, forest managers, policymakers, and climate experts are keenly interested in utilizing an evidence-based approach to map forest fire susceptibility (Ghorbanzadeh et al., 2019). This approach helps explain the devastating impacts of fires (Cascio, 2018) and provides insights into prospective situations in environments susceptible to forest fires (Boubeta et al., 2015). In contemporary times, such information is crucial as it helps mitigate the adverse consequences on potentially affected groups and supports conservation and restoration efforts (Oliveira et al., 2012). Additionally, this data assists relevant agencies in allocating resources, implementing early warning systems, coordinating emergency services, and developing effective forest management and planning strategies (Bui et al., 2017). Therefore, accurate forest fire susceptibility prediction and fire mitigation enhancement plans are prioritized to advocate and encourage sustainable development.

Researchers have developed various prediction models that focus on exploring the spatiotemporal patterns linking various elements (such as topography, climate, biophysics, and disturbances) for instigating a forest fire (Boubeta et al., 2015). Recently, Geographical Information System (GIS) techniques have proven valuable in manipulating spatial information and contributing to generating forest fire susceptibility maps (Duarte and Teodoro, 2016). Several studies have utilized geospatial databases and techniques to optimize prescribed fire allocations and assess forest fire risk mapping (Pradhan et al., 2007). These strategies enhance management plans, mitigate fire spread, and decrease the frequency of fires.

Various modeling approaches have been envisaged to explore the intricacy of forest fires. Researchers have classified fire models into three categories: (i) Physically-based models, which demand precise and comprehensive data to predict the spatiotemporal patterns of forest fires accurately; (ii) Semi-empirical models, which rely on a series of laboratory experiments for their formulation; and (iii) Empirical modeling, which employs statistical and data mining techniques to analyze fire-related variables for prediction (Hong et al., 2018). Empirical modeling revolves around the regional environmental element analysis influencing forest fire ignition and spread. It underscores the importance of each factor in predicting the relevant variables. This approach makes it possible to forecast future fires and identify risk zones (Gigović et al., 2019). Empirical modeling is categorized into various types based on the specific approaches used and the nature of the analyzed forest fire data. Basic modeling starts with multiple criteria decision analysis like the analytic hierarchy process used by Sivrikaya and Küçük (2022). Some major types of empirical modeling are Regression Models, where statistical regression techniques establish relationships between predictor variables and the response variable (e.g., fire occurrence and fire size). They are widely used in analysing fire behavior and risk factors (Pourghasemi, 2016; Tien Bui et al., 2016). Time series modeling analyses data collected over time to identify patterns and trends. This type of empirical modeling can help understand fire occurrences and spread

over different periods (Huesca et al., 2009). Data mining techniques are another empirical model employed to extract valuable information from large datasets, enabling the identification of hidden patterns and relationships between variables related to forest fires (Pourtaghi et al., 2016). Simulation models are used to recreate fire behavior and spread under various artificially set conditions. They help researchers and fire managers understand how different factors influence fire dynamics (Kanga and Singh, 2017). Recent advancements in machine learning models (MLM) make them capable of handling complex datasets and can offer valuable insights into forest fire dynamics. Some of the works, like Gaudreau et al. (2016) have taken one step forward by modeling the wildfire in future climatic scenarios. They are proven to be reliable and efficient means of determining the relative critical attributes from various fire datasets (Sayad et al., 2019). In recent times, support vector machines (Jaafari and Pourghasemi, 2019), neural networks (Li et al., 2020), kernel logistic regression are benchmark classifiers (Tien Bui et al., 2016), and hybrid artificial methods (Jaafari et al., 2019) like random forest (Bustillo Sánchez et al., 2021; Cao et al., 2017) have gained popularity in fire prediction studies. Other emerging machine learning models are also used globally to predict forest fire risk zonation (Abid, 2021). However, each MLM has several pros and cons; therefore, model ensembling is the second step in improving the predictive ability of forest fire susceptibility in large-scale studies (Gigović et al., 2019; Tuyen et al., 2021). Ensemble approaches are often considered better than single models because they improve predictive performance, reduce overfitting, and provide more robust and reliable results (Ganaie et al., 2022).

This study focuses on the NEI due to its significance in forest fire occurrences. Approximately half of India's forest fires occur in this specific region (Bahuguna and Singh, 2002). Considering the severity of forest fire incidents and their associated vulnerability, conducting a detailed analysis of forest fire occurrence and assessing the region's fire vulnerability is of utmost importance. Small-scale susceptibility modeling exercises at the local level (Babu et al., 2023) and state level (Das et al., 2023) have been recently successful elsewhere in India. However, our study stands out as the first comprehensive investigation in India, covering a larger important geographic area that is highly prone to forest fire. This research aims to address the gaps in the comprehensive study of forest fire susceptibility modeling in NEI by utilizing multiple machine learning models viz., Boosted Regression Tree (BRT), Random Forest (RF), Support Vector Machine (SVM), Classification Regression Tree (CART), and Multivariate Adaptive Regression Splines (MARS) as these models have gained popularity in a wide range of natural hazards modeling (Jain et al., 2020). BRT and RF were selected for their ability to handle complex interactions and high-dimensional data, while SVM offers robustness to outliers and non-linear patterns. CART and MARS provide interpretability and flexibility, vital for understanding and modeling the diverse factors influencing forest fire susceptibility in the region. Finally, ensembling them with a recently improved modeling methodological framework will help identify zones susceptible to forest fires and offer practical solutions to minimize the risk of such incidents in the region. The findings of this study can contribute to the development of early warning systems, resource allocation for fire arrest, and the formulation of effective forest fire prevention and management plans. Hence, the study aims to (1) delineate forest fire susceptibility zones in NEI employing ML modeling techniques and an ensemble method and (2) offer critical insights to enhance the efficacy of forest fire management strategies in the region.

This comprehensive approach to modeling forest fire susceptibility not only integrates multiple recently improved methodologies but also emphasizes their applicability across vast geographic regions. By harnessing the strengths of various modeling techniques, this new approach offers a more robust and reliable analysis, a rarity in forest fire susceptibility studies. The research extends beyond regional boundaries to encompass large geographical areas in the NEI. This detailed analysis enables policymakers and stakeholders to pinpoint areas of high

susceptibility within each state of NEI, enabling the implementation of targeted management strategies. Moreover, the methodology employed, which integrates satellite imagery, GIS techniques, and improved modeling, can be replicated elsewhere, ensuring efficient forest fire management on a regional to large scale globally.

2. Material and methods

2.1. Study area

NEI region, comprising eight northeastern states (viz., Assam, Arunachal Pradesh, Meghalaya, Manipur, Nagaland, Mizoram, Sikkim, and Tripura) and northern part of West Bengal of India, bounded by latitudes 21.94° – 29.46° N and longitudes 87.98° – 97.41° E, is one of the 36 biodiversity hotspots of the world covering a vast area of 2,73,304 km² (See Fig. 1). The region is known for its breathtaking natural beauty, rich cultural heritage, and ethnic diversity. The climate is a dynamic blend of humid subtropical characteristics, featuring a combination of hot and humid summers, intense monsoons, and relatively mild winters. The region boasts an impressive annual average rainfall of 2068 mm, ranking among the highest precipitation-receiving areas worldwide. Here, one can find *Cherrapunji* (or *Mawsynram*), an astonishing location recognized as the wettest place on Earth, receiving approximately 12,000 mm of rainfall annually (Jain et al., 2013). The average temperature in the area fluctuates between 5 °C and 30 °C, while the mean relative humidity typically ranges from 70% to 85% throughout the year. (Jhajharia et al., 2009). It is home to various indigenous communities with unique traditions, languages, and customs. The region is home to numerous endemic species of flora and fauna, including rare and endangered ones. In addition to paddy cultivation, the NEI thrives on tea production along with cash crops like bamboo, various fruit crops, and orchids, contributing significantly to its agricultural economy (Jhajharia et al., 2012). The region is also famous for its diverse wildlife, making it a popular tourist destination. NEI boasts diverse forest types, including tropical deciduous, and evergreen to alpine forest (Roy et al., 2015; Tripathi et al., 2016). The region is particularly renowned for its extensive bamboo mixed forests (Basumatary et al., 2015). Despite

accounting for 23.75% of India's total forest cover (ISFR, 2021), the northeastern states have been experiencing a consistent decline in forested areas (ISFR, 2019). Between 2011 and 2019, excluding Assam, six regional states witnessed a substantial reduction of approximately 18% in forest cover, resulting in a loss of nearly 25,012 km² of forests in the preceding decade (ISFR, 2021). This region faces significant economic challenges, with forests bearing the brunt of anthropogenic pressures, primarily stemming from traditional shifting cultivation practices known as “*jhum*” (Tripathi et al., 2016).

2.2. Forest fire and background locations

Forest fire location data ($n = 32,816$) between 2018 and 2019 from the Suomi National Polar-orbiting Partnership - Visible Infrared Imaging Radiometer Suite (SNPP-VIIRS) sensor with a spatial resolution of 375 m were downloaded from the Forest Survey of India (FSI - fast 3.0, 2019). SNPP-VIIRS data are usually ground verified and stored in the fast 3.0 database of FSI on time. Due to the extensive size of the landscape area, conducting independent field verification of the fire data becomes a cumbersome and time-consuming process. To ensure accuracy of the fire data, a sample of 109 locations was randomly selected across Arunachal Pradesh, Nagaland, North Bengal, and Mizoram. These locations were field-verified by regional stakeholders in May 2023. The results revealed that 97% of the fire incidents matched precisely with the observations on the ground or indicated a history of forest fire occurrence since 2018. To address spatial sampling biases and spatial autocorrelation challenges in the analysis, we employed a bootstrapping approach using the *spThin* package in R (Aiello-Lammens et al., 2019). To achieve a 1 km spatial resolution for the modeling exercise, we conducted 50 iterations of bootstrapping using a minimum thinning distance of 1 km. This process resulted in obtaining 12,577 unique locations. This approach effectively minimized the removal of records while significantly mitigating sampling bias (Aiello-Lammens et al., 2015; Radosavljevic and Anderson, 2014). Spatially-constrained and combined methods for generating background points show greater efficacy in modeling outcomes despite ongoing debates regarding the optimal method for generating them (Shipley et al., 2022). For this modeling exercise, we used Arc Map 10.3 (ESRI, Redlands, CA), employing a combined method to create a 5 km buffer around the chosen forest fire locations, generating 12,577 background points. These background points comprised 50% of randomly sampled points within the buffer and the remaining 50% within the study region but fell outside the buffer region. The Spatial Autocorrelation (Global Moran's I) was assessed using ArcMap 10.3 (ESRI, Redlands, CA) to determine whether the pattern exhibited is clustered, dispersed, or random.

2.3. Selection of predictor variable

Initially, we selected 32 predictors based on an extensive literature review and consultation with regional experts, as these factors were identified to have a direct or indirect influence, either positively or negatively, on forest fire occurrence in the region (Guo et al., 2020; Table 1, Table S1). All predictor variables were categorized into four major types, namely topographic ($n = 5$), climatic ($n = 9$), anthropogenic ($n = 4$), and biophysical ($n = 14$), and initially tested pairwise with the Pearson correlation coefficient test and some variables were found to be correlated (Fig. S1). However, high correlation among predictors can negatively impact the model's performance and cause over-estimation in model outcome (Graham, 2003). Consequently, we used the variance inflation factor (VIF) analysis, more precisely the *vifstep* (step function) and *vifcor* (correlation function) provided by the *usdm* package in R, in order to identify multicollinearity among the chosen predictors (Naimi, 2015). For calculating VIF and tolerance (TOL) values, these functions utilized a stepwise procedure to find and eliminate very collinear variables. If a variable's VIF is greater than ten and its TOL is less than one, it is frequently deemed inappropriate to include it in the

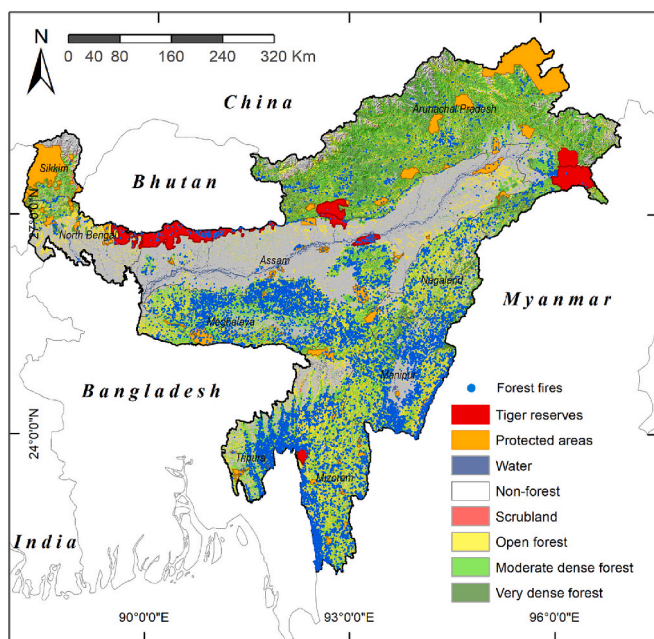


Fig. 1. Geographic coverage of the study area depicting the NEI region, which comprises of eight northeastern states (viz., Assam, Arunachal Pradesh, Meghalaya, Manipur, Nagaland, Mizoram, Sikkim, and Tripura) and the northern part of West Bengal of India.

Table 1

The predictor variables used in model building were selected after conducting a multicollinearity test.

Categories	Predictors (abbreviations)	Explanations
Topographic	Topographic Position Index (tpi; Fig. S2)	The Topographic Position Index (TPI) is derived from a Digital Elevation Model (DEM) by calculating the difference between a cell's elevation value and its neighboring cells' average elevation. Positive TPI values indicate that the cell is situated at a higher elevation than its surroundings, whereas negative values indicate a lower elevation (Weisse and Goldman, 2017). Areas with higher TPI values, such as ridge tops, tend to have increased exposure to wind and are more susceptible to fire propagation. In contrast, areas with lower TPI values, such as depressions, may have higher fuel moisture content and lower fire susceptibility (Evers et al., 2022).
	Slope (Degree; slope; Fig. S3)	The slope serves as an indicator of the elevation's rate of change and the terrain's steepness. It influences both the speed and trajectory of fire propagation. Fires tend to advance more swiftly when moving uphill compared to downhill, meaning that the steeper the slope, the greater the fire's velocity (Baltaci and Yildirim, 2020; Kushla and Ripple, 1997; Rothermel, 1991). The slope layer was derived from the ASTER Digital Elevation Model (Yamaguchi et al., 1998).
	Distance to nearest water bodies (km; water_dis; Fig. S4)	Forest fire occurrence is more likely to increase with greater distance from water bodies. The data was obtained from Open Street Map (OSM), and the Euclidean distance was calculated using ArcMap 10.
	Topographic Wetness Index (twi; Fig. S5)	The Topographic Wetness Index (TWI), also known as the Compound Topographic Index (CTI), is developed by the interplay of the upstream contributing area and the slope of the terrain. This index serves as an indicator of water accumulation tendencies across a slope. It highlights areas with a high potential for soil moisture (represented by high values) and areas that dry up rapidly (represented by low values), thus serving as reliable surrogates of forest fire occurrence (Krueger et al., 2017). The TWI layer was derived from the ASTER Digital Elevation Model.
Climatic	Precipitation of warmest Quarter (mm; bio_18; Fig. S6)	Higher values indicate that increased moisture content is a negative indicator for the spread of fires. The precipitation data is averaged from the warmest months. The data was retrieved from the Worldclim version 2.1 (Fick and Hijmans, 2017; Kalabokidis et al., 2013) at 30 arc sec. Spatial resolution (~1km ²).
	Solar Radiation (KJ/ m ² /day; mn_srad; Fig. S7)	Solar radiation is a significant factor in causing wildfires in eastern India (Das et al., 2023). The average Solar Radiation was calculated from the database provided by WorldClim Version-2 of 30 arc-seconds resolution (Fick and Hijmans, 2017).
	Wind Speed (m/ s; wind_mn; Fig. S8)	Wind Speed is a significant factor in causing forest fires (Jahdi et al., 2014). The average Wind Speed was calculated from the database provided by WorldClim Version-2 of 30 arc-seconds resolution (Fick and Hijmans, 2017).
Anthropogenic	Human Footprint (hu_foot; Fig. S9)	The Global Human Footprint Dataset of the Last of the Wild Project, Version 2 (Center for International Earth Science Information Network - CIESIN - Columbia University, 2018) is the Human Influence Index (HII) was used as a predictor for forest fire occurrence as Human-induced alterations to the environment have emerged as a primary catalyst for the escalation of forest fire occurrences (Harvey, 2016).
	Human Population Density (hu_pop; Fig. S10)	Increasing forest dependency among rural people indicates demographic pressure on forests, hence heightening the likelihood of forest fires (Kolanek et al., 2021). The Gridded Population of the World v4 (Center for International Earth Science Information Network - CIESIN - Columbia University, 2018) was used to create Human Population density/ km ² .
	Distance to Nearest Roads and Rails (rl_road_dis; Fig. S11)	The likelihood of forest fire occurrence tends to rise in correlation with the extent of the Anthropogenic disturbance. Open Street Map (OSM) was the data source for road and rail lines, which are linear infrastructure, and the Euclidean distance was computed using ArcMap 10.
	Forest to Human Population Ratio (for_pop_ratio; Fig. S12)	Increasing forest dependency among rural people indicates demographic pressure on forests, hence heightening the likelihood of forest fires (Kolanek et al., 2021). The forest cover map derived by Hansen et al., 2013 computed the percentage of its coverage per Km ² , then the Gridded Population of the World v4 (Center for International Earth Science Information Network - CIESIN - Columbia University, 2018) was used to create the Forest to Human Population Ratio.
Biophysical	% Evergreen Forest (prop_evrgreen; Fig. S13)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Evergreen Forest patches in the NEI were extracted, and the percentage of its coverage per Km ² was computed. Different forest types have different types of impacts on forest fires (Silva et al., 2009)
	% Semi-Evergreen Forest (prp_smi_ever; Fig. S14)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Semi-Evergreen Forest patches in the NEI were extracted, and the percentage of its coverage per Km ² was computed. Different forest types have different types of impacts on forest fire (Silva et al., 2009)
	% Moist Deciduous Forest (prp_mst_d; Fig. S15)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Moist Deciduous Forest patches in the NEI were extracted, and the percentage of its coverage per Km ² was computed. Different forest types have different types of impacts on forest fires (Silva et al., 2009)
	% Forest cover/ Km ² (prcnt_forst; Fig. S16)	The forest cover map was derived from Hansen et al., 2013 and computed the percentage of its coverage per Km ² . Different forest types have different types of impacts on forest fire (Silva et al., 2009)
	% Bamboo Forest (prop_bamb; Fig. S17)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Bamboo Forest patches in the NEI were extracted, and the percentage of its coverage per Km ² was computed.
	% Jhum Cultivation (prop_jhum; Fig. S18)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Jhum cultivated areas in the NEI were extracted, and the percentage of its coverage per Km ² was computed.
	% Subtropical Broadleaved Forest (prp_sub_brd; Fig. S19)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Subtropical Broadleaved Forest in the NEI and the percentage of its coverage per Km ² were extracted.
	% Moist Temperate Forest (prp_mst_t; Fig. S20)	The study utilized a comprehensive vegetation-type map of India (1:50,000 scale) developed by Roy et al. (2015). The Moist Temperate Forest in the NEI was extracted, and the percentage of its coverage per Km ² was computed.
	Enhanced Vegetation Index (evi; Fig. S21)	The Enhanced Vegetation Index (EVI) has numerous benefits due to its insensitivity to atmospheric interference, ability to adapt to different surface coverings, and applicability in dense vegetation and cloud cover areas. The EVI index quantifies vegetation changes and correlates with fire recurrence (Das et al., 2023). MODIS data product id. MOD13A1 with 500 m spatial resolution was used.
	Normalized Difference Water Index (ndwi; Fig. S22)	Decreasing NDWI values show higher susceptibility to forest fire (Das et al., 2023). MODIS data product id. MCD43A4 with 500 m spatial resolution was used.
	Forest Patch Area (Fragmentation; area_mn; Fig. S23)	Forest Patch Area is a fragmentation index created using the forest cover map derived from Hansen et al., 2013. Studies show deforestation-induced fragmentation increases forest fire incidents (Silva Junior et al., 2018). The Forest Patch Area Index was calculated with moving window analysis in FRAGSTATS v4 (McGarigal et al., 2023).
	Perimeter To Area Ratio (para; Fig. S24)	The forest patch perimeter to area ratio is a fragmentation index using a forest cover map derived from Hansen et al. (2013). Studies show deforestation-induced fragmentation increases forest fire incidents (Silva Junior et al., 2018). The forest perimeter to area ratio was calculated by implying moving window analysis in FRAGSTATS v4 (McGarigal et al., 2023).

established model (Iban and Sekertekin, 2022). Additionally, we performed pairwise calculations of Pearson's correlation coefficient among the selected variables, ensuring that the correlation coefficient remained within the range of ± 0.75 for any combination of variable pairs (Tehrany et al., 2019). Table 1 comprehensively describes the selected predictors, including their data source, data processing method, importance as predictors for forest fire susceptibility, and corresponding abbreviations. Further, a description of all excluded predictors is presented in Table S1.

2.4. Model building and validations

The Forest Fire Susceptibility Modeling (FFSM) plays a crucial role in the overall modeling process. It involves utilizing Machine Learning Techniques (MLTs) to analyze the data on forest fire occurrences and predictor variables. The objective is to accurately forecast the likelihood of forest fires transpiring in the study area. To estimate the probability of forest fires transpiring in the NEI region, we employed the 'sdm 1.1–8' package in R (Naimi and Araújo, 2016). An ensemble platform specifically created for species distribution modeling (SDM) is provided by this R package. However, for this study, the machine learning models shared a common objective with species distribution models (SDMs): pinpoint areas with a heightened susceptibility to forest fires (Babu et al., 2023; Barros-Rosa et al., 2022). Given the ongoing debate surrounding the selection of the optimal algorithm for environmental modeling, we chose to construct models utilizing five distinct machine learning classification algorithms (MLMs), i.e., Boosted Regression Tree (BRT; Friedman, 2001), Random Forests (RF; Breiman, 2001), Support Vector Machine (SVM; Vapnik, 1999), Classification and Regression Trees (CART; Breiman, 1984) and, Multivariate Adaptive Regression Spline (MARS; Friedman, 1991); all renowned for their robust predictive capabilities, and modeled ensemble integrated into the 'sdm 1.1–8' package. This approach was taken to explore various methodologies and compare their effectiveness in the context of this research. These modeling techniques are described in detail and have real-world applications in Pourghasemi et al. (2020) and Jain et al. (2020). To mitigate modeling overestimation and prevent bias, we implemented several strategies. Firstly, we employed thinned forest fire locations and background points generated through a combination of spatially-constrained and combined methods. Additionally, to ensure the accuracy of our models, we utilized a subset of 23 predictors out of the original 32, which were selected based on a VIF analysis. By implementing these pre-model building techniques, we ensured the enhancement of the reliability and robustness of the modeling outcomes.

By default, we utilized the recommended parameters for all MLMs, except for Random Forest (RF), where we expressly set the number of trees to 500 as an additional parameter. A higher number of trees helps reduce the resulting model's bias. We calibrated and validated each model after fitting each MLM with optimized parameters using a 10-fold cross-validation (CV) design. For each run of the cross-validation procedure, we reserved 30% of the forest fire occurrence data as a validation set, with the remaining 70% of the data being used to train the model. This rigorous validation methodology was adopted following the method explained by Eskandari et al. (2021). The approach generated 50 resulting models, consisting of 10 models for each machine learning algorithm.

Validation of MLMs is a critical step in assessing each model's predictive capabilities, as Mohammadi et al. (2014) emphasized. To evaluate the accuracy and reliability of each MLM, we employed robust statistical measures provided by the 'sdm 1.1–8' package. These are kappa (Allouche et al., 2006), true skill statistics (TSS; Allouche et al., 2006), and area under the receiver operator characteristics (ROC) curve (AUC; Hanley and McNeil, 1982). Model complexity was evaluated for each case by computing the difference between the AUC (train) and AUC

(CV), represented as Δ AUC (Jarnevich et al., 2018). The Δ AUC scores from each model offer insights into their sensitivity to the training data (Jarnevich et al., 2018). Moreover, the Kruskal-Wallis test was conducted to determine whether a significant difference in prediction accuracy exists among the classifiers.

2.5. Ensemble model

Ensemble modeling capitalizes on the strengths of multiple models to enhance predictive accuracy, increase stability, improve generalization, provide robustness against overfitting, and increase model interpretability (Ganaie et al., 2022). Combining the predictions of diverse models, ensemble modeling achieves more accurate and reliable results by mitigating the risks of errors and biases, resulting in stable predictions that generalize well to unseen data (Ganaie et al., 2022). All the fitted models obtained from the five ML algorithms exhibited favorable performance measures. The AUC for all models surpassed 0.75, while the TSS exceeded 0.45. We used these results to build the final ensemble model, employing the AUC-weighted ensemble method (Achu et al., 2021; Tehrany et al., 2019). This technique utilizes the performance metrics to effectively combine the predictions from all models, ensuring a robust and accurate final model. However, for visualization of models derived from each ML algorithm, we ensemble each of the ten models using the AUC weighted ensemble technique. The cumulative response curves for understanding the association between predictors and the response variable for each MLM and ensemble model were prepared using the *rcurve* function of 'sdm 1.1–8' package of R.

2.6. Variable importance

We calculated the relative importance of different variables ($n = 23$) in the ensemble model using training data and assessed them using correlation-based (cor) and area under the receiver operating characteristic curve (AUC) methods. Correlation-based variable importance calculates the correlation coefficient between the predictor and response, with higher absolute values indicating greater importance. In contrast, AUC-based variable importance evaluates the predictive power of each predictor variable by measuring its contribution to the overall model performance (Strobl et al., 2007). The *getVarImp* function of 'sdm 1.1–8' package of R was used to generate bar plots indicating the level of importance in the ensemble model.

2.7. Binarize ensemble model

To assess the management implications of the forest fire susceptible map generated using the ensemble approach, we first applied a mask to the continuous-scale ensemble forest fire map using the forest cover data of the region. Subsequently, we conducted a Welch two-sample one-tailed *t*-test to determine if the mean forest fire probabilities within the forested areas of the region were significantly higher compared to the forest fire probabilities outside the forested areas. Secondly, we identified a threshold for classifying the continuous-scale forest fire probabilities within the forested region into two categories: forest fire susceptible and non-susceptible zones. To accomplish this, we employed the maximizes (sensitivity + specificity)/2 method implemented in the *PresenceAbsence* package (Freeman and Freeman, 2012) in R. This approach aims to optimize the balance between sensitivity (the ability to identify susceptible areas correctly) and specificity (the ability to identify non-susceptible areas correctly) to determine an appropriate threshold for classifying the forest fire susceptible map of NEI. Further state-wise forest fire susceptible areas were computed in ArcMap 10.3. All the step-by-step modeling methods employed in the study are represented in the schematic illustration presented in Fig. 2.

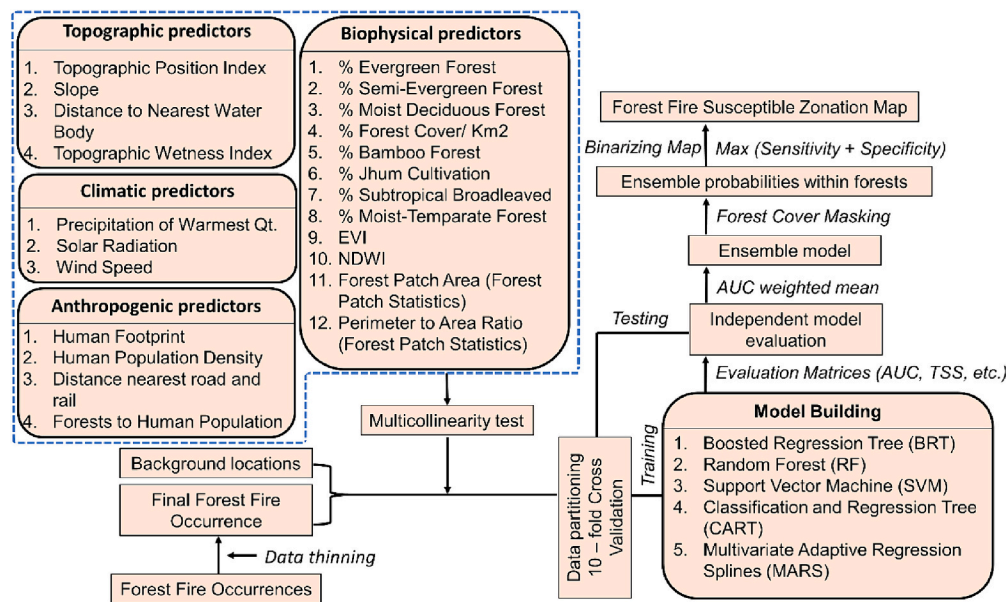


Fig. 2. Schematic illustration of the analytical framework for deriving the forest fire susceptible zonation of NEI.

3. Results

3.1. Pre-modeling exercise

The forest fire occurrence inventory data for the entire NEI was compiled for 2018 and 2019, utilizing data from the FSI fast 3.0 database. Notably, the period between February and April exhibited a significant rise in forest fire incidents for both years. Specifically, during March, the highest number of forest fires occurred compared to any month of the year, with the least fire events occurring during July and August. The forest fire locations selected for modeling exhibited a modest level of spatial autocorrelation, displaying a propensity towards clustering (Moran's $I = 0.03$, $P \leq 0.05$). Following an exhaustive literature review and considering the regional context, a meticulous selection of forest fire predictor variables ($n = 32$) was conducted. These variables predominantly fell within four categories: topographic, climatic, anthropogenic, and bio-physical (refer to Table 1 and Table S1). Variables with high VIF and TOL values were identified and removed from the analysis to avoid multicollinearity issues. Ultimately, only variables ($n = 23$; Table 2) within the optimal VIF and TOL values level were retained for further analysis. Before conducting the multicollinearity test, the selection of predictor variables underwent an additional cross-check using the Pearson correlation test. This test revealed that certain variables exhibited high correlation, with Pearson correlation coefficients exceeding ± 0.75 (see Fig. S1). However, after

Table 2
Multicollinearity statistics of selected predictor variables.

Predictors	Collinearity Statistics		Predictors	Collinearity Statistics	
	VIF	Tolerance		VIF	Tolerance
tpi	1.18	0.85	prop_evrgreen	1.24	0.80
slope	2.87	0.35	prp_smi_ever	1.52	0.66
water_dis	1.18	0.85	prp_mst_d	2.03	0.49
twi	2.55	0.39	prcnt_forst	4.93	0.20
bio_18	1.43	0.70	prop_bamb	1.31	0.76
mn_srad	2.41	0.42	prop_jhum	1.83	0.55
wind_mn	3.06	0.33	prp_sub_brd	2.28	0.44
hu_foot	3.42	0.29	prp_mst_t	2.35	0.43
hu_pop	2.10	0.48	evi	4.53	0.22
rl_road_dis	1.75	0.57	ndwi	1.54	0.65
for_pop_ratio	1.22	0.82	area_mn	6.94	0.14
			para	4.40	0.23

eliminating these highly correlated variables through the multicollinearity test, the pairwise Pearson correlation coefficients for the selected predictors were all below ± 0.75 (see Fig. 3). This process effectively mitigated the potential impact of bias and overestimation on the resulting models, enhancing the reliability and accuracy of the models.

3.2. Model performance MLMs comparisons

The performance of each MLM was assessed using several metrics, such as AUC, TSS, Kappa, and phi. The outcomes of these evaluations are presented in Table 3. Additionally, Fig. 4 visually illustrates the predictive performance of each model with AUC values for both training and test datasets. Upon analysing the four performance metrics, it was evident that the selected models exhibited high consistency, except for the CART. The performance of the remaining models displayed only slight variations within a small range, as shown in Table 3. Additionally, there was no discernible variation in the classifiers' prediction accuracy according to the Kruskal-Wallis statistic ($p > 0.05$).

Among the five MLMs evaluated, the AUC analysis revealed that RF and SVM performed remarkably well with training AUC values of 0.868 ± 0.002 and 0.857 ± 0.001 , respectively. MARS followed this, and BRT gained training AUC scores of 0.829 ± 0.001 and 0.809 ± 0.002 , respectively, while CART exhibited poor performance with a low training AUC of 0.781 ± 0.001 . Similar trends were observed for TSS, where RF, SVM, MARS, BRT outperform CART (Table 3). All models exhibited robust agreement with the predicted results in terms of accuracy. Additionally, when considering the kappa and phi for recognizing accurate forest fire pixels, all models, RF, SVM, MARS, BRT outperformed CART (Table 3). The Δ AUC was lowest for CART with 0.002, followed by MARS and BRT with both values of 0.003. The Δ AUC was highest for RF and SVM with the values of 0.022 and 0.021, respectively, indicating the sensitivity of the data used for fitted models. Among all model runs RF and SVM retained all the provided predictors (Fig. S35, Fig. S36), followed by MARS ($n = 13$; Fig. S38), and only three predictors were contributed in BRT and CART (Fig. S34, Fig. S37). However, the AUC-weighted ensemble model yields considerable results with other stand-alone MLMs (AUC = 0.840; TSS = 0.53; kappa = 0.536; phi = 0.55, sensitivity = 0.900; and specificity = 0.673). While the AUC value for the ensemble model is slightly lower compared to the RF and SVM, its sensitivity and specificity values surpass those of any other

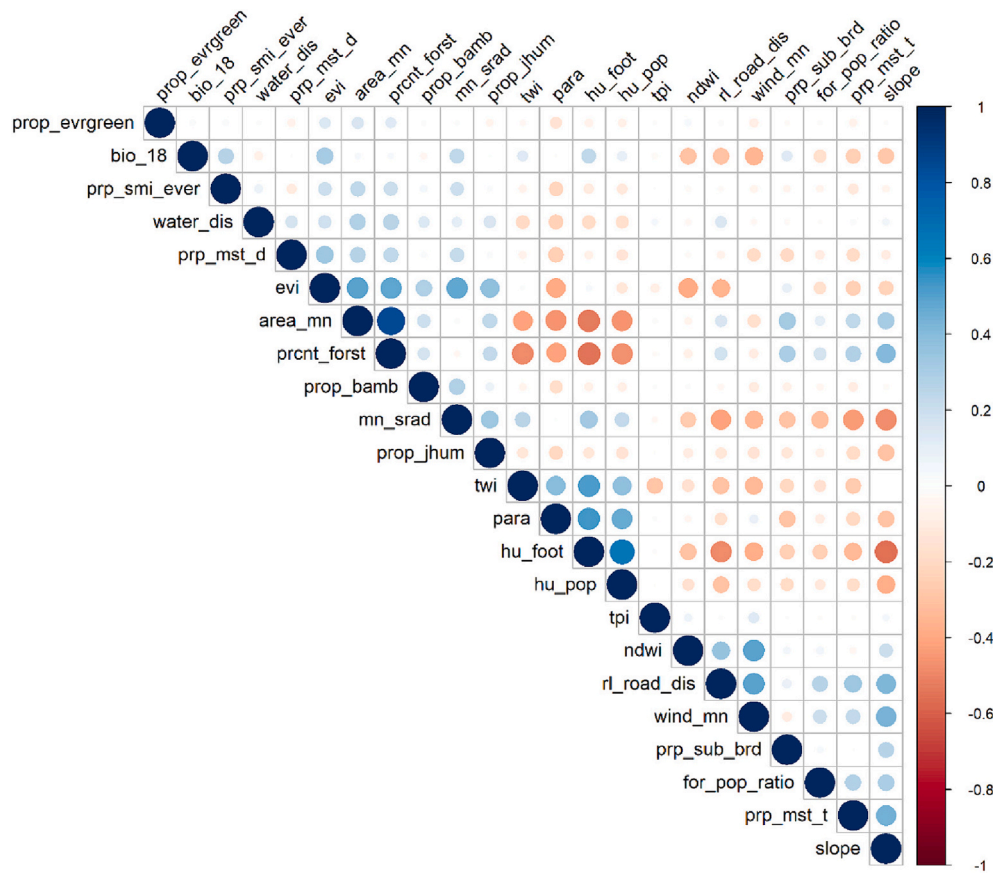


Fig. 3. The Pearson correlation matrix for the chosen twenty-three predictor variables revealed correlation coefficients of $\leq +0.75$ and ≥ -0.75 among the predictors, which is why these variables were selected for modeling.

Table 3
Model evaluation metrics for each participating forest fire susceptible model.

Models	Forest Fire Data	AUC	Δ AUC	TSS	Kappa	phi	Sensitivity	Specificity
BRT	Training	0.809	0.003	0.512	0.513	0.519	0.834	0.679
	CV	0.806		0.513	0.514	0.521	0.842	0.671
RF	Training	0.868	0.022	0.584	0.585	0.591	0.861	0.723
	CV	0.846		0.560	0.562	0.579	0.898	0.662
SVM	Training	0.857	0.021	0.566	0.568	0.581	0.890	0.676
	CV	0.836		0.554	0.556	0.574	0.904	0.650
CART	Training	0.781	0.002	0.493	0.494	0.502	0.835	0.658
	CV	0.779		0.492	0.493	0.498	0.820	0.671
MARS	Training	0.829	0.003	0.539	0.541	0.557	0.890	0.650
	CV	0.826		0.542	0.544	0.559	0.888	0.654

stand-alone MLMs. This indicates that the ensemble model correctly identifies positive forest fire cases and accurately predicts negative cases compared to the individual MLMs.

3.3. Predictor variable importance and response curves

Among all the predictors involved, *evi*, *mn_srad*, and *hu-pop* (Fig. S34 – S38) were unanimously chosen by all five modeling methods due to their significant contributions to the ensemble model. According to the correlation metric, their average contributions are 19.9%, 17.6%, and 10% respectively (Fig. 6A). The frequency of forest fire occurrences in the research area is perfectly positively correlated with the *evi* and *mn_srad* response curves. In contrast, for *hu-pop*, the response curve shows a sharp decline with low human population density but a monotonic increase in fire probability with increasing human population density. The variable *prop_jhum* holds significance as it was selected by RF, SVM, and MARS, (Fig. S35–36, 38), contributing 8.6% to the

ensemble model according to the correlation metric (Fig. 6A). It shows a minor influence on the increase in forest fire probability in the landscape with the increment of percentage of *jhum* cultivated area/ Km² (Fig. 7). Two other variables, namely *ndwi* and *for_pop_ratio*, contributed 2.1% and 1.7% to the ensemble model (Fig. 6A). These variables demonstrate a negative relationship with forest fire probability (Fig. 7). Moreover, the remaining predictors listed in Table 1 made minor contributions to the ensemble model, ranging from 0.3% to 1.2%, based on the results obtained from the correlation metric (Fig. 6A). The variable contribution calculated using the AUC metric also aligns with the trends of variable importance observed through the correlation metric (Fig. 6B).

3.4. Forest fire maps and fire susceptible zonation

The spatial distributions of the susceptibility classes produced by the ensemble model and each MLM are shown in Fig. 5. The forest fire probability zones from the ensemble model were defined from a

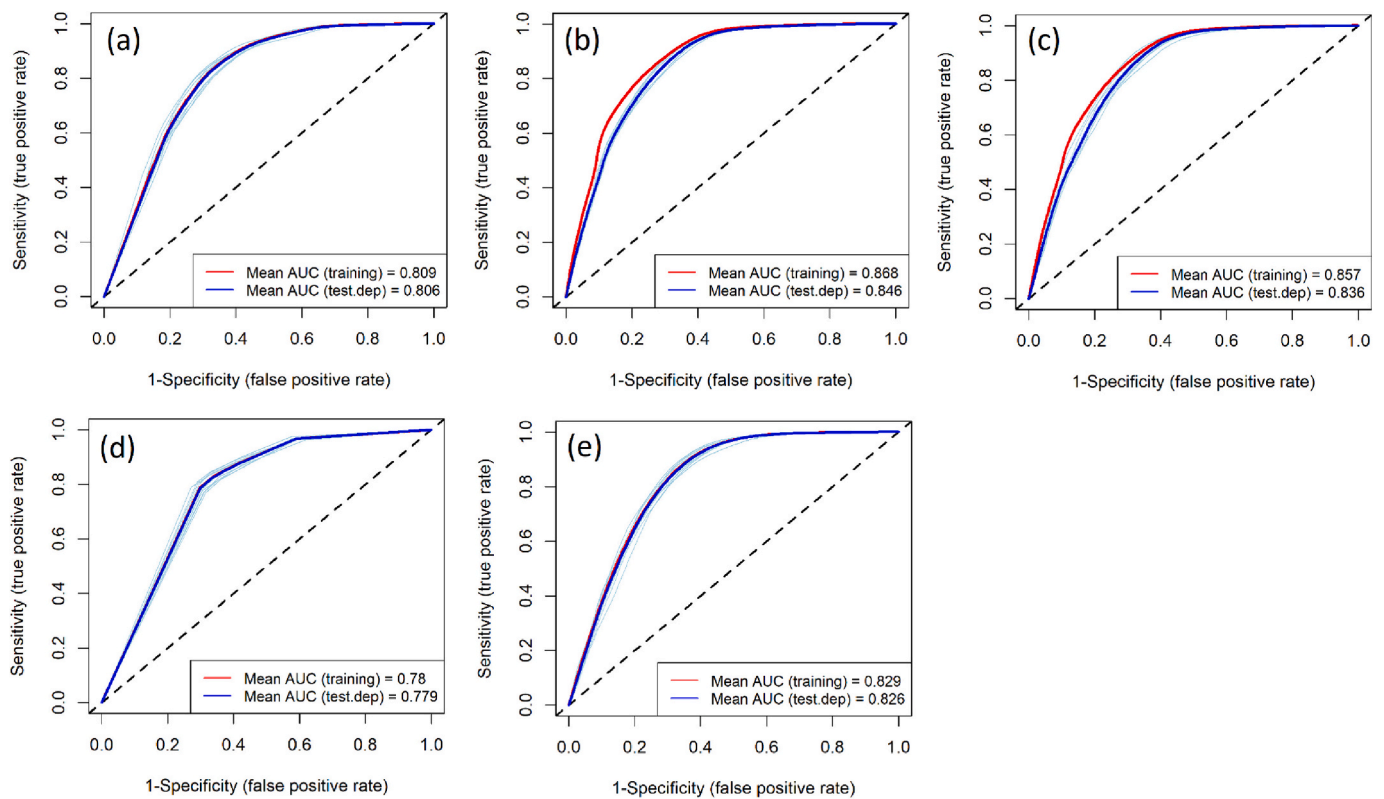


Fig. 4. ROC curves and the average AUC of the participating models were analyzed. K-fold cross-validation ($k = 10$) was employed to estimate and compare the performance of various machine learning models. Following ten rounds of training/validation rotations for each model, the average AUC was computed [(a) BRT, (b) RF, (c) SVM, (d) CART, and (e) MARS model].

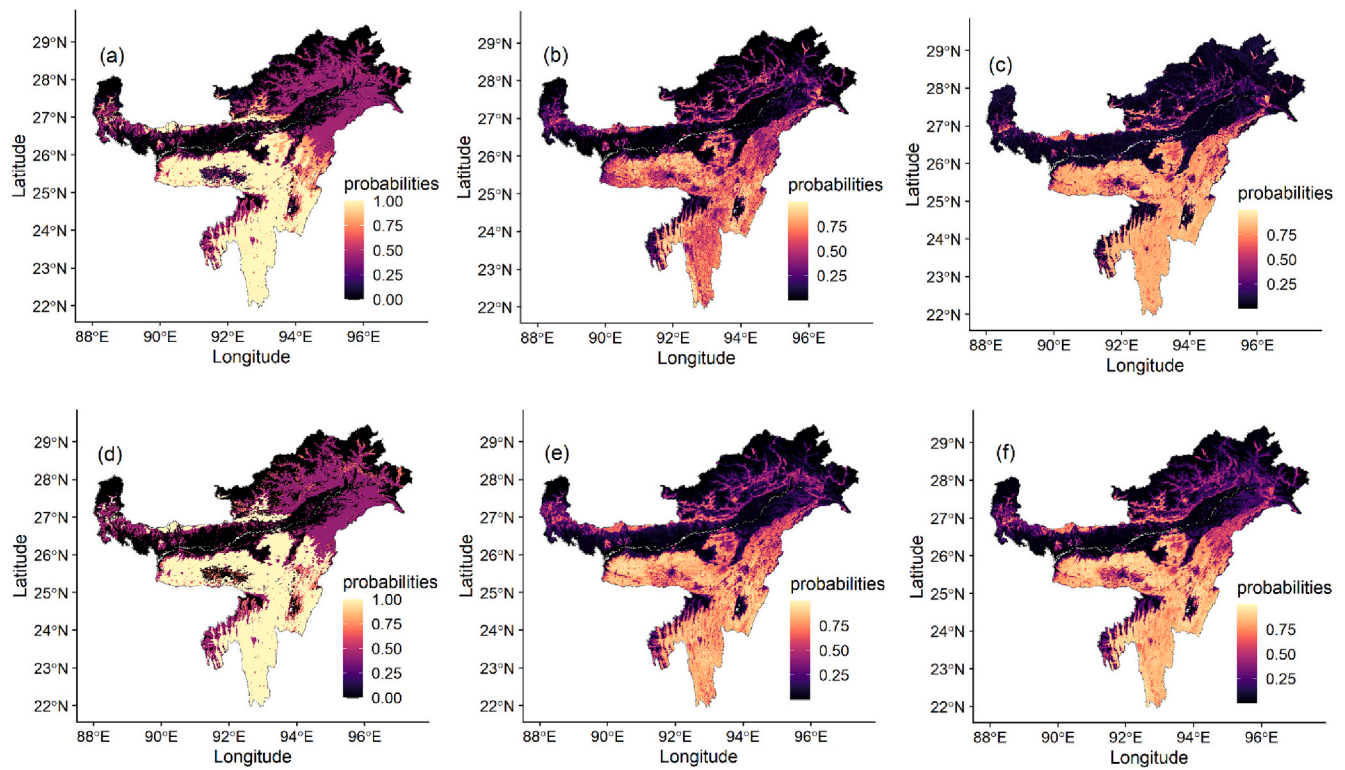


Fig. 5. The predicted probability surface forest fire susceptibility maps of stand-alone MLMs and their corresponding AUC-weighted ensemble model. (a) BRT, (b) RF, (c) SVM, (d) CART, (e) MARS, and (f) Ensemble model.

Table 4

The table presents forest susceptibility statistics for different states in NEI, along with their respective geographic areas and the availability of different types of forests ("Survey of India updated administrative boundary"; ^a forest area derived from Hansen et al., 2013).

Sl. No.	State	Geographical Area ^a (GA; Km ²)	Total Forest area ^b (TFA; Km ²)	% of GA	Forest Fire Susceptible Area (FFSA; km ²)	% of GA = (FFSA/GA) *100	% of TFA = (FFSA/TFA) *100
1	Arunachal Pradesh	82,128.19	64,772.34	78.87	2099	2.556	3.24
2	Assam	78,413.20	32,536.36	41.49	15,853	20.217	48.72
3	Manipur	22,331.58	17,814.68	79.77	16,340	73.170	91.72
4	Meghalaya	22,418.07	18,147.26	80.95	14,902	66.473	82.12
5	Mizoram	21,066.75	19,752.36	93.76	18,806	89.269	95.21
6	Nagaland	16,585.85	13,677.44	82.46	7773	46.865	56.83
7	Sikkim	7129.75	2802.21	39.30	35	0.491	1.25
8	Tripura	10,444.24	7678.90	73.52	4625	44.283	60.23
9	Northern West Bengal	12,786.20	4588.21	35.88	302	2.362	6.58

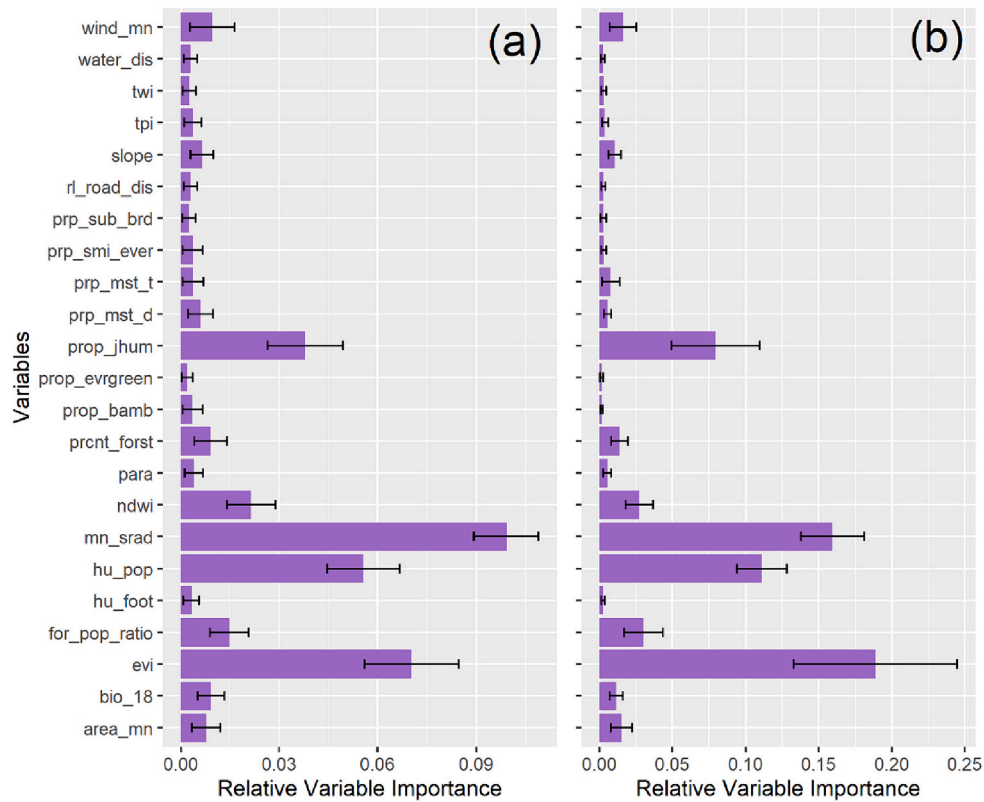


Fig. 6. 23 predictors' variable importance on the likelihood of forest fires occurring, as determined by the ensemble model using the (a) correlation and the (b) AUC-based methods. The error bar shows the highest and lowest important values over ten validation iterations in each independent model, while the purple bar shows the mean permutation importance. For descriptions of the independent models and variable acronyms, see Table 1 and Fig. 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

management perspective. It is obtained by a two-step process. First, the forest fire probability zone within the forested region was masked by the forest cover of which 1,74,011.71 km² area falls within the forested region and 99,292.29 km² area is non-forested region (Table 4). However, A Welch Two Sample one-tailed *t*-test revealed that the mean forest fire probability is significantly higher in the forested region compared to the non-forest region (Fig. S39; *t* = 247.61, *p* < 0.005). In the second step, the forest fire probability distributions within the forested region were binarized using a threshold value of 0.61, which was determined using the maximize (Sensitivity + specificity)/2 method. This resulted in identifying the forest fire risk zone for NEI, covering an area of 80,735 km², where probability values ranging from 0.61 to 1.00 best explain the forest fire risk. Fig. 8 illustrates the spatial distribution of final forest fire probabilities represented using a color gradient covering 29.54% of the

total geographic and 44.416% of the forested area of NEI (Table 4). The varying degrees of forest susceptibility in different states of NEI are observed. Mizoram has the highest susceptibility, with 89.27% of its geographic area being at risk, followed by Manipur (73.17%), Meghalaya (66.47%), Nagaland (46.86%), Tripura (44.28%), Assam (20.22%), Arunachal Pradesh (2.56%), and Northern West Bengal (2.36%). Sikkim has the lowest vulnerability with only 0.491% of its geographic area being susceptible to forest fires (Table 4).

4. Discussion

Every year, forest fires in Eastern India consistently destroy large tracts of forest, endangering human health, and the environment (Wang et al., 2021). Over the past two decades, numerous efforts have been

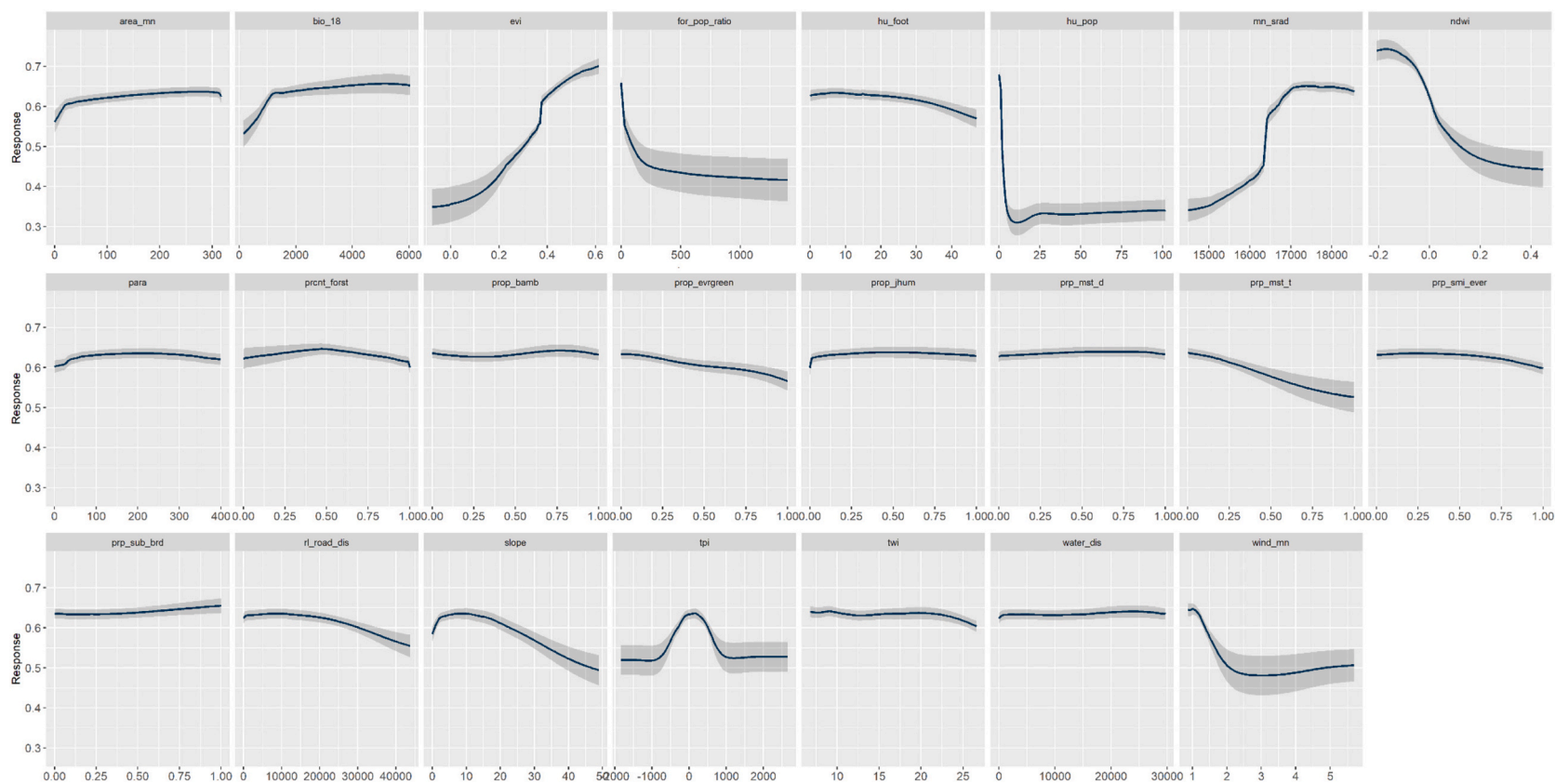


Fig. 7. Response curves illustrating the relationship between predictor factors and the probability of forest fire events, are produced by the ensemble model. The rug plots give an illustration of how different gradients of predictor factors affect the frequency of fires.

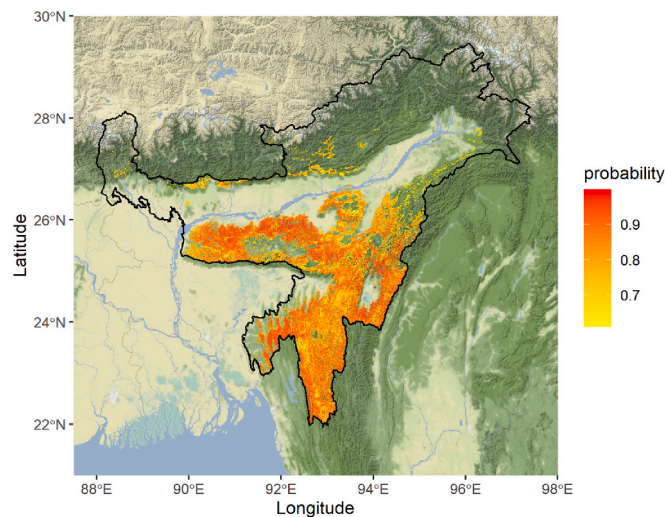


Fig. 8. The ensemble model produced a map showing the probability of forest fires in different regions of NEI. The map displays fire probability values exceeding the calculated probability thresholds using the maximize (sensitivity + specificity)/2 method. The color gradient, ranging from yellow to red, illustrates the increasing fire probability within the computed forest fire susceptible zone of NEI. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

made to assess forest fire vulnerability in the region, including studies using weighted multi-criteria analysis with various factors affecting fire behavior in 2014 (Chakraborty et al., 2014). Similar methodologies have been employed in state-level studies in several states of NEI, leading to the development of forest fire susceptibility maps (Gupta and Shukla, 2022; Pramanick et al., 2023; Dhar et al., 2023; Lamat et al., 2021; Bar et al., 2022; Ahmad et al., 2018; Jayasri et al., 2016; Laha et al., 2021; Naskar et al., 2022; Mao and Gogoi, 2010, etc.). The impact of forest fire on aerosol concentration (Badarinath et al., 2007) and pollution by emission (Bar et al., 2022) were conducted in the region. However, the application of improved modeling techniques using MLMs on a large scale has been relatively unexplored not only in the NEI but also in other parts of the country, except for a few state-level studies in Mizoram, Odisha, Uttarakhand, and Himachal Pradesh (Bera et al., 2022; Bhattacharjee et al., 2023; Das et al., 2023; Shahfahad et al., 2022) and at a regional scale in areas like Ayodhya hill, West Bengal, and the Shettihalli landscape in Shivamogga district located in central Western Ghats (Babu et al., 2023; Saha et al., 2023). In the light of the escalating severity of forest fire incidences in NEI, our study represents the first-ever endeavour in the country to assess forest fire susceptibility over a vast geographic area of 2,73,304 km² employing five improved Machine Learning Models (MLMs) and ensembling techniques. Our forest fire occurrence inventory data for the years 2018 and 2019 revealed a significant rise in forest fire incidents between February and April, with March having the highest number of fire events that followed similar trends explained by Gupta and Shukla (2022). In our study, RF and SVM demonstrated remarkable performance, outperforming the other models which is also reflected in the work of Thach et al. (2018) while analysing forest fire severity at Thuan Chau area. RF and SVM's ability to handle complexity, capture nonlinear relationships, mitigate overfitting, and provide insights into feature importance make them particularly well-suited for identifying forest fire susceptible regions in the NEI compared to other models. However, the ensemble model showed considerable results, excelling in accurately identifying positive forest fire cases and predicting negative cases compared to individual MLMs. Past research has also proven that model ensembling approaches are preferable to individual model use due to their ability to combine multiple models, improving accuracy and generalization while mitigating risks associated with individual model weaknesses. By integrating

diverse modeling perspectives and spreading the risk of model failure, ensembling enhances robustness and flexibility in predictive modeling tasks (Babu et al., 2023). All models unanimously chose the Enhanced Vegetation Index (EVI), mean solar radiation, and human population density for their significant contributions. EVI is designed to be highly responsive to variations in vegetation health and density, indicating potential fuel load and flammability in the forest. These characteristics are pivotal factors in determining the level of fire susceptibility in the study area. The human population density is a surrogate for anthropogenic pressure on the natural ecosystem that triggers forest fire occurrences. In similar works using different models, Bjånes et al. (2021) and Kantarcioglu et al. (2023) suggested that anthropogenic factors are predisposing factors for ignition in forest. An earlier study in the region also showed that proximity to road settlements helped to demarcate the fire-prone zones (Chakraborty et al., 2014). During the summer months, intense solar heat leads to drying up of vegetation, especially in the south and southeast regions, creating ample fuel for natural forest fires. Das et al., 2023 also mentioned forest fires triggered by solar heat in Odisha. The triggering of forest fires by solar heat is considered a potential effect of climate change. As global temperatures rise due to climate change, the increased solar heat can exacerbate vegetation drying, making forests more susceptible to wildfires (Macias Fauria et al., 2011).

Shifting cultivation is an agricultural practice where a patch of land is cleared in different parts of NEI using the slash-and-burn method. Different crops are grown in the cleared area for a few seasons before the farmers rotate to a new patch of land (Ramakrishnan, 1984). During shifting cultivation, the cleared land is intensively cultivated for two to three years, after which it is left to undergo natural forest regeneration, known as the fallow period. The length of the fallow period, ranging from 3 to 15 years, depends on land availability and population size. Once the fallow period ends, the vegetation is slashed and burned again, commencing a new shifting cultivation cycle (Tripathi and Barik, 2003). However, the increasing frequency of intensive shifting cultivation and reduced fallow periods, also known as the *jhum* cycle, poses severe threats to ecosystems nowadays. Such practices have led to declining soil fertility, reduced crop yields, and increased soil erosion, endangering the overall environmental health (Saha et al., 2012). Our findings demonstrate the impact of forest burning caused by human-made forest fires. Specifically, we observed that the percentage of shifting cultivated area had a significant contribution to the occurrence of forest fires in specific parts of the landscape. The data showed that the amount of forested area available to human population per unit area played a significant role in contributing to forest fires. This indicates that areas where the human population relies more heavily on forests are more prone to forest fires. For example, in the northern part of Arunachal Pradesh, where the human density is considerably lower compared to Mizoram, Manipur, and Nagaland, the availability of forest land for indigenous communities per unit area is much higher. Interestingly, in these regions with greater forest availability for the local population, the incidence of forest fires is significantly lower compared to other parts of the landscape. While this proposition requires more in-depth research, people in regions experiencing high pressure on forests due to indigenous activities could consider adopting alternative livelihoods. By doing so, they can reduce their dependence on forest resources and decrease the incidence of forest fires in such areas. Further study is needed to explore the feasibility and potential impact of implementing these alternative livelihood strategies.

NDWI (Normalized Difference Water Index) is another natural factor that negatively influences forest fires. To counteract this effect, the government should devise conservation strategies for preserving natural springs and preventing the drying up of natural water channels. These measures would contribute to reducing the likelihood and intensity of forest fires.

Overall, the forest fire probability zones within the forested region of NEI covering an area of 80,735 km², best explained the forest fire risk.

The resulting forest fire susceptibility map covers 29.54% of the total geographic and 44.42% of the forested area. Annual average (2001–2020) forest fire burn area over the Indian Himalayan Region shows similar trends of forest fire susceptibility derived from our study (Bar et al., 2022).

Our study revealed varying degrees of forest susceptibility across states in NEI. Mizoram exhibited the highest susceptibility at 89.27%. Other regional, remote sensing-based studies (Bhattacharjee et al., 2023) and indexes of forest fire risks (Pramanick et al., 2023) also revealed a similar trend. Time-series-based research using the Google Earth engine has demonstrated that Mizoram is highly vulnerable to forest fires and shows alarming trends of forest fire occurrence require urgent attention to combat its effect on forests and people (Gupta and Shukla, 2022). Manipur ranks as the second-highest forest-susceptible zone, encompassing 73.17% of the total geographic area of the state. A study employing weighted sum modeling and ISODATA clustering for visual interpretation also corroborated this pattern, indicating a high susceptibility to forest fires in the region (Puri et al., 2011). The 3rd largest forest-susceptible area found in Meghalaya covers 66.47% of the state area. A similar agreement with the spatial distribution could be drawn from two other studies in the state (Dhar et al., 2023) and the Ribhoi region of Meghalaya (Lamat et al., 2021), where human habitation, road networks, climate, elevation, slope, aspect, forest type, soil moisture, forest canopy density, and wind significantly contributed to forest fire in the region. Nagaland accounts for 46.865% of land susceptible to forest fires. An intriguing study conducted in the region highlights the role of invasive dwarf bamboo, rapidly colonizing and altering the forest structure and diversity in the Dzukou Valley as a trigger for forest fires (Mao and Gogoi, 2010). Our study also reveals that *jhum* cultivation is another contributing factor to forest fires, which is consistent with previous research findings (Padalia and Mondal, 2014). In an effort to detect the Forest Burnt Area, RISAT-1 SAR Data was utilized for pre- and post-fire events in Tripura (Jayasri et al., 2016). Our study revealed that 44.28% of the geographic area of Tripura is considered susceptible to forest fires. Among all the sister states of NEI, Assam has a relatively smaller forest area despite its large geographic extent, making up 20.22% of the forest fire susceptible zone. Arunachal Pradesh has a relatively smaller forest fire susceptible zone, accounting for only 2.56% of its extensive geographic area. An interesting study revealed a high correlation with socio-economic factors, as 42.3% of forest fire locations were concentrated in areas with high poverty indexes (Ahmad et al., 2018). Additionally, the emerging concept of citizen science in forest fire management was explored in the state by developing a Mobile App to gather data (Qayum et al., 2020). Until now, no study has been conducted in the North Bengal area. However, our study is the first to report a very minimal forest fire susceptible area, primarily within the Buxa Tiger Reserve, accounting for only 2.36% of the total geographic area. According to our study, Sikkim exhibited the lowest vulnerability, with only 0.49% of its geographic area being susceptible to forest fires. Our research findings are consistent with other studies in Sikkim, which also indicate that the lower part of the state is prone to forest fires (Laha et al., 2021; Naskar et al., 2022; Sharma et al., 2014).

5. Conclusions

The developed methodology, integrating satellite imagery, GIS techniques, and advanced machine learning models, offers a replicable framework applicable to diverse ecosystems globally. By demonstrating its efficacy in NEI, this research provides a blueprint for conducting forest fire susceptibility assessments tailored to specific environmental and socio-economic contexts. Forest fires pose a global concern, impacting ecosystems, economies, and communities. Effective management necessitates proactive measures, including prevention, early detection, and mitigation strategies. By identifying susceptibility zones and understanding underlying drivers, this study empowers

policymakers to develop evidence-based management plans, reducing ecological damage and safeguarding biodiversity and livelihoods. Incorporating climatic variables into susceptibility models addresses the nexus between climate change and forest fires, informing adaptation strategies and resilience-building efforts. Furthermore, it underscores the importance of preserving forests as carbon sinks and reducing emissions from forest fires. Forest fires profoundly affect biodiversity, ecosystem services, and ecological balance. By pinpointing susceptibility zones, the research informs conservation efforts and guides sustainable land management strategies that reconcile conservation goals with socio-economic development objectives. Given that forest fires transcend national boundaries, the findings and methodologies presented facilitate global collaboration and knowledge exchange among stakeholders. By sharing best practices and innovative approaches, the global community can collectively enhance its capacity to address the multifaceted challenges posed by forest fires.

While this study provides valuable insights into forest fire susceptibility in North-East India (NEI), it's essential to acknowledge potential limitations for future research. One notable limitation is the reliance on data from the SNPP – VIIRS sensor for the years 2018–2019, which might not capture the full range of temporal variations in forest fire dynamics in the region. To address this, future studies could consider incorporating longer-term temporal datasets to identify temporal dynamics of forest fire incidences at a large spatial scale. This will reflect the progress of regional field level forest fire mitigation strategies and help in identifying the local best management practices for controlling forest fires and help in replicating those practices at a large scale with a broader goal of reducing forest fire incidences in the region.

Moreover, the elimination of nine crucial predictor variables during multicollinearity analysis may have impacted the assessment of forest fires in the region. Subsequent research could delve into advanced statistical techniques or alternative methods to address correlated predictors more adeptly, guaranteeing a comprehensive grasp of the elements influencing forest fire susceptibility. The current investigation exclusively concentrated on a solitary spatial scale (1 km) for pinpointing factors influencing forest fire occurrence in the region. Future advanced modeling studies could gain from exploring the incorporation of multi-scale analysis, not only to accurately pinpoint influential predictor variables but also to identify the scale at which they exert the most influence on forest fires in the region.

The moderate spatial autocorrelation observed in the used forest fire locations can be accounted for in future forest fire susceptibility models. Spatial lag is a concept commonly used in spatial analysis to account for spatial autocorrelation. It represents a weighted average of a variable across neighboring locations. Including spatial lag as a predictor variable in machine learning models used for forest fire susceptibility mapping can help capture spatial autocorrelation patterns and improve model performance (Liu et al., 2022).

However, this study provides insightful information for implementing effective forest fire management plans in NEI. The ensemble model's superiority in correctly identifying forest fire cases and accurately predicting non-fire cases showcases its potential as a robust forest fire risk assessment tool. The identified vulnerable zones can guide policymakers and forest managers in adopting targeted strategies to mitigate the impacts of forest fires and safeguard the region's forests. Supplementary Material - 2 contains policy recommendations at the local level, derived from this study, for the district-level regional management of forest fires within each state in the study area. Near real-time fire location data obtained from MODIS/FIRMS can be utilized to generate fire alerts, serving as an early warning system in identified forest fire risk areas (Chakraborty et al., 2014). To build a resilient community against forest fires, it is crucial to adopt innovative approaches in forest fire management to address current issues and potential future challenges within the identified vulnerable regions. Implementing strategies that enable observation, analysis, amalgamation, storage, and dissemination of forest fire incidents to various stakeholders for timely decision-making is

paramount (Joshi et al., 2018). Lack of awareness among indigenous people, inadequate training in fire management, infrastructure, and technology, as well as a lack of policy and funding, make it difficult to adapt as well as respond to the fires that occur in the identified regions that are growing in size and frequency. Additionally, the scarcity of scientific studies to comprehend forest fire dynamics in the eastern Himalayan region further compounds the challenges. Therefore, funding such studies would have a significant impact on mitigating forest fires and improving the overall situation in the area (Wang et al., 2021). Furthermore, the methodology used in this study, incorporating satellite imagery, GIS techniques, and modeling algorithms, can be applied in other geographical regions worldwide to enhance forest fire management efforts. Overall, this research contributes to understanding forest fire susceptibility in NEI and lays the foundation for proactive and informed forest fire management practices.

CRedit authorship contribution statement

Mriganka Shekhar Sarkar: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Writing – original draft, writing management recommendation in the supplementary material. **Bishal Kumar Majhi:** Conceptualization, Data curation, Software, Formal analysis, Writing – review & editing, writing management recommendation in supplementary material. **Bhawna Pathak:** Data curation, Formal analysis, Writing – review & editing. **Tridipa Biswas:** Data curation, Writing – review & editing, writing management recommendation in supplementary material. **Soumik Mahapatra:** Writing management recommendation in supplementary material. **Devendra Kumar:** Writing – review & editing. **Indra D. Bhatt:** Conceptualization, Project administration, Funding acquisition, Supervision, Writing – review & editing. **Jagadish C. Kuniyal:** Conceptualization, Project administration, Funding acquisition, Supervision, Writing – review & editing. **Sunil Nautiyal:** Conceptualization, Investigation, Supervision, Writing – review & editing, funding acquisition, writing management recommendation in supplementary material.

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Declaration of generative AI in scientific writing

During the preparation of this work, the authors used ChatGPT 3.5 (<https://chat.openai.com>) partially in order to improve the English language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The article utilized data from open-source repositories. The data acquisition links are specified in the manuscript (refer FSI; fast 3.0, 2019 for forest fire locations and Table 1 for predictor variables). Additionally, the processed data will be provided upon reasonable request and R codes employed in the study are provided in the supplementary material.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoinf.2024.102598>.

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